

EE102

Lab 1: Introduction to Digital Oscilloscopes

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Purpose

The purpose of this lab is to students used to the lab equipment such as oscilloscope, breadboard and other basic electronic circuit elements. In the first 5 step of this lab, the focus is mostly on oscilloscope using, its functions and their details. It is aimed that students can learn graphing and digital visualization of analog electronic components. Additionally, in the last step, it is targeted that students to obtain basic electronic knowledge. By designing a circuit on a breadboard with resistor and capacitance and displaying voltage levels and measuring the time delay and phase angle by using oscilloscope is expected.

Design Specification

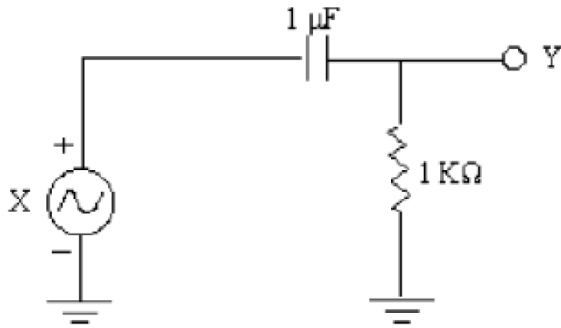


Figure 1: Circuit design used in step 6

This schema is given in the report. Students only rearranged this circuit on the breadboard using 2 probes connected to CH1 (X) and CH2 (Y) of the oscilloscope, 1 KΩ resistor, 1 μF capacitor and a couple of jumper wires.

More detail about this circuit can be found under 6th step.

Lab Work

(Methodology and Results parts are combined under this title be clearer.)

1. **Use the compensation signal of the oscilloscope and compensate your probes as explained in the above-mentioned document. Explain in your report how you compensated your probes.**

Firstly, oscilloscope is opened and connected the probe to CH1 input with BNC cable.



Figure 2: Probe connection to CH 1

After that, to make precise measurements, I compensated the oscilloscope by turning the small slit on the probe, shown in the photo below, with small flat screwdriver which is given with the probe. The point is make the signals as rectangular as possible.

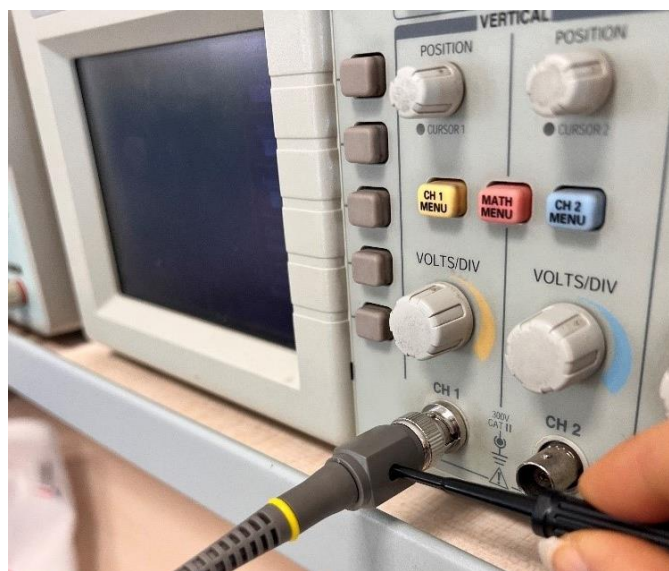


Figure 3: Compensation of the oscilloscope

There are three conditions of signals in terms of compensation: undercompensated, properly compensated and overcompensated. Under-compensated and overcompensated conditions are problematic because they deceive with wrong and inconsequent results during experiment.

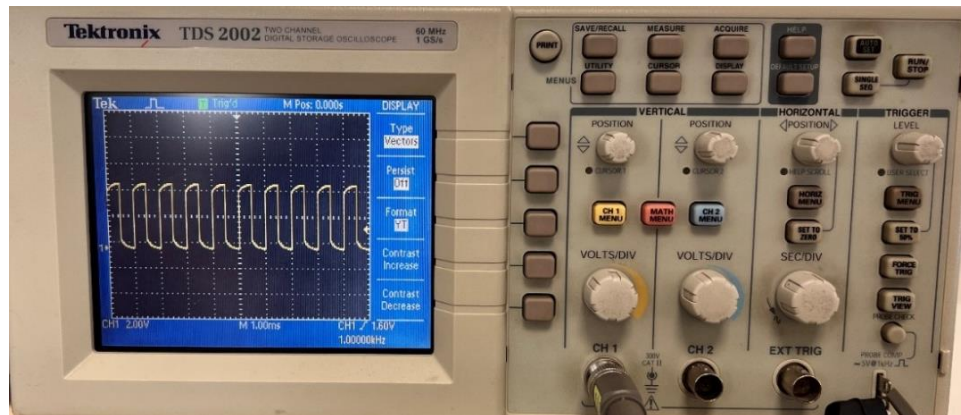


Figure 4: Undercompensated

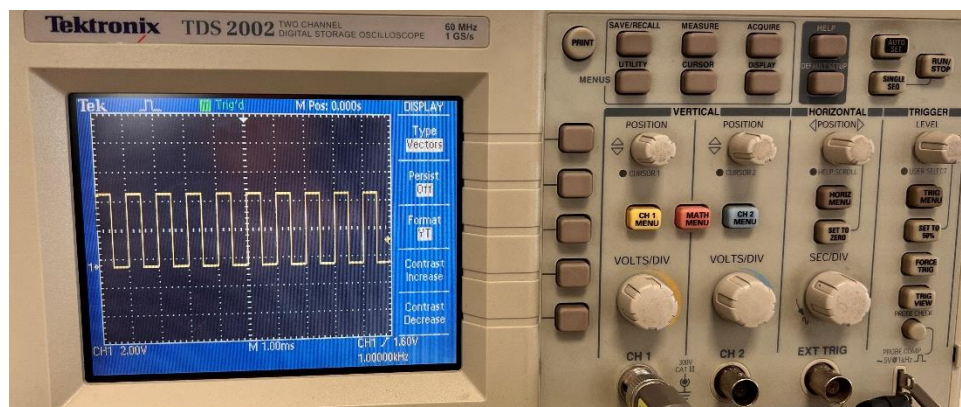


Figure 5: Properly compensated

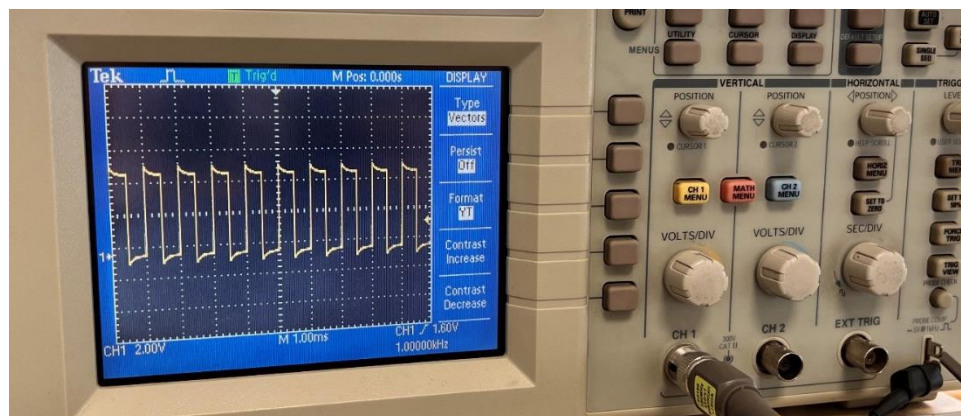


Figure 6: Overcompensated

2. Using a signal generator, apply a 5 Vpp sinusoidal signal with frequency 1 kHz. Use the oscilloscope to monitor this signal. First, use positive edge triggering and report what you see on the screen. Then, apply negative edge triggering and report the result.

To begin with, I adjusted the amplitude 2.5 Vpp and frequency 1000 kHz on the signal generator which is connected to the oscilloscope.

(Note: Since I have forgotten to take photos of this step, I had to retake these later. And since I could not go to the school due to current situations, I have taken these photos in my father's office's lab. That's why oscilloscope and signal generator is different in this step. Also in lab, the amplitude given to signal generator is doubled on the oscilloscope but in this devices, it is not. Therefore, it is what is given in this step and in further steps it will be half of desired amplitude.)

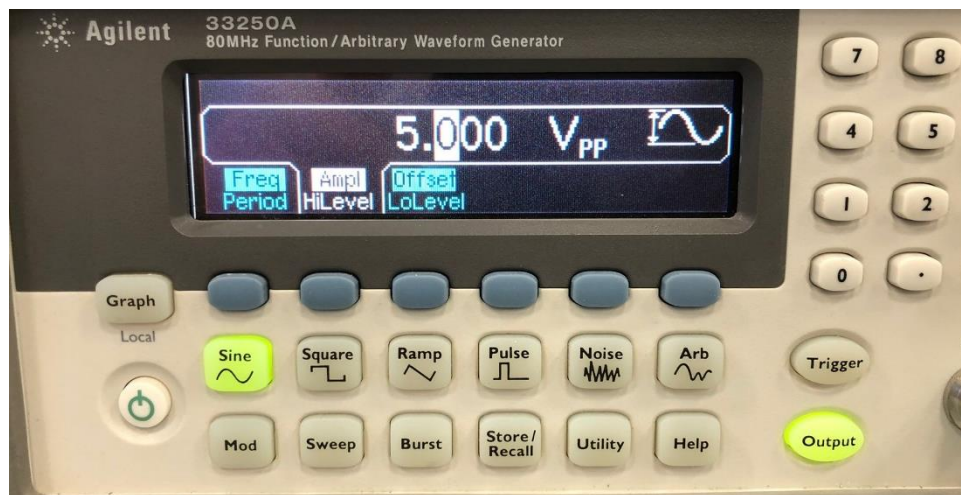


Figure 7: Amplitude adjustment, 5 Vpp signals will be created



Figure 8: Frequency adjustment

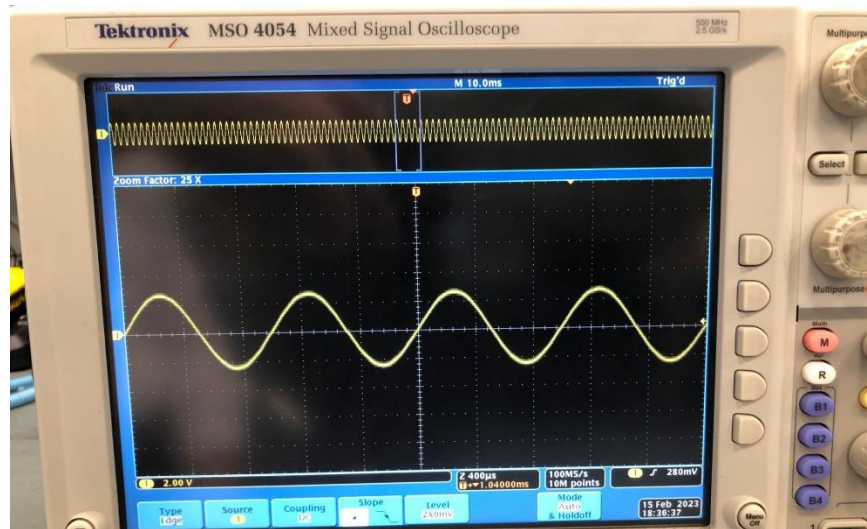


Figure 9: Positive edge triggering

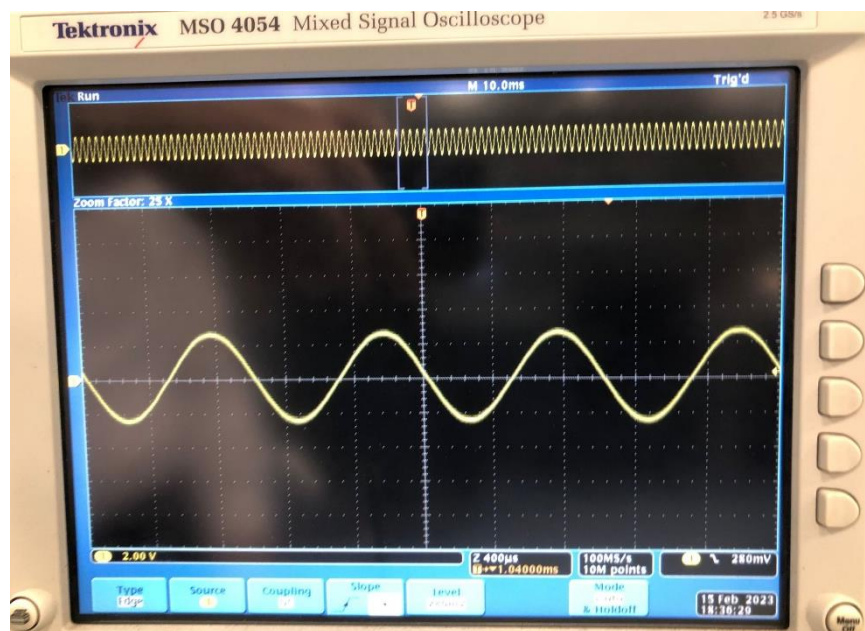


Figure 10: Negative edge triggering

We can see that these displays are precise because they are triggered. It can be seen from the small green “Trig’d” located at the top of the screen. It means that the triggering adjustment is made and graphs are proper.

In the third photo, positive edge triggering is shown. Starting from the column indicated by dashed lines, graph starts to rise. On the contrary, in the final photo, it starts to fall down from indicated line which is negative edge triggering.

3. Apply a 1 Vpp triangular wave with 2 kHz frequency to the oscilloscope. Observe and report the effect of turning the trigger knob on the oscilloscope. Comment in your report on what the triggering concept is.



Figure 11: Amplitude adjustment, 1 Vpp signals will be created

The reason why the amplitude is half of what is wanted is signal generator is not terminated by 50 Ω and in this case, output voltage can be twice the given value. According to the measurements, it is so. Therefore, half of the desired amplitude is used in signal generator in this experiment.

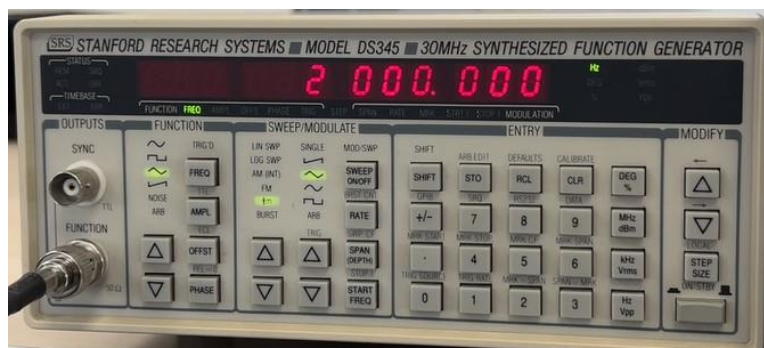


Figure 12: Frequency adjustment



Figure 13: Trigger knob

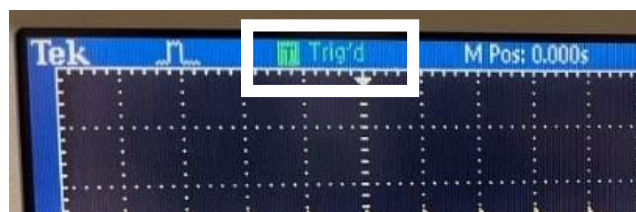


Figure 14: "Trig'd" symbol, showing whether graph is triggered or not

Triggering is adjusted by turning the trigger knob. Now, in the photos below it can be seen what graph looks like when it is not triggered. Photos can be low quality because they are taken from a video to catch the fast change in displays. Also, whether the graph is triggered or not can be checked from “Trig’d” word. Non-triggered photos do not include this while the middle triggered one does.

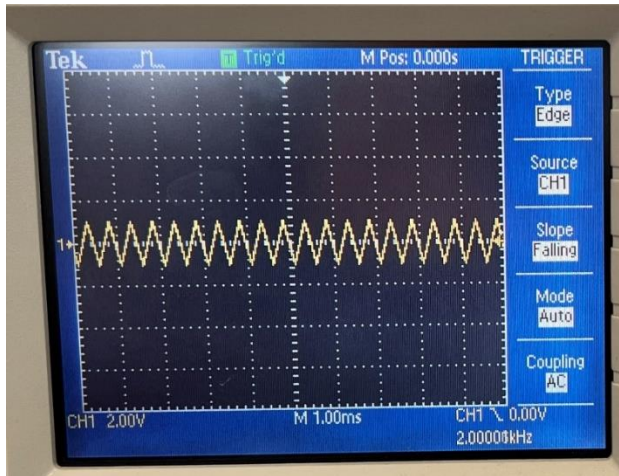


Figure 15: Triggered graph containing "Trig'd" symbol

This is the normal version of applied signals. Photos below displaying a the process of turning the trigger knob.

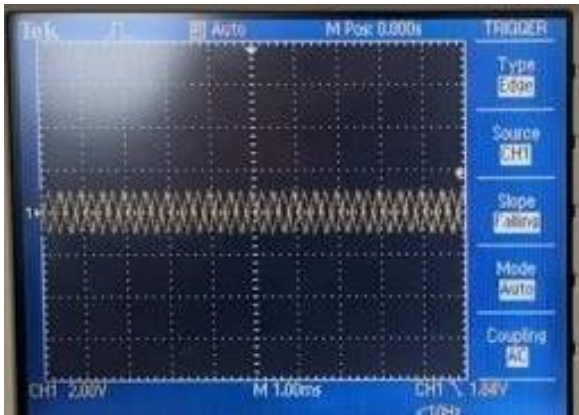


Figure 16: Triggering step 1

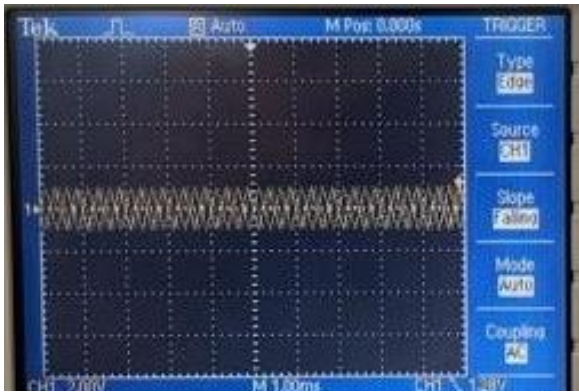


Figure 17: Triggering step 2

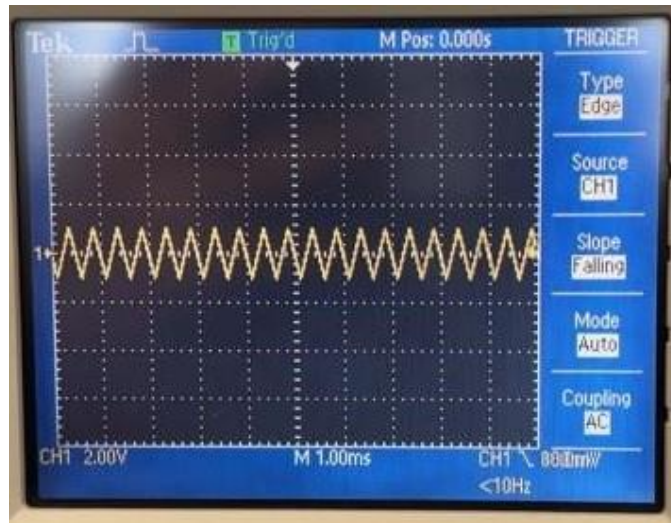


Figure 18: Triggering step 3, triggered graph

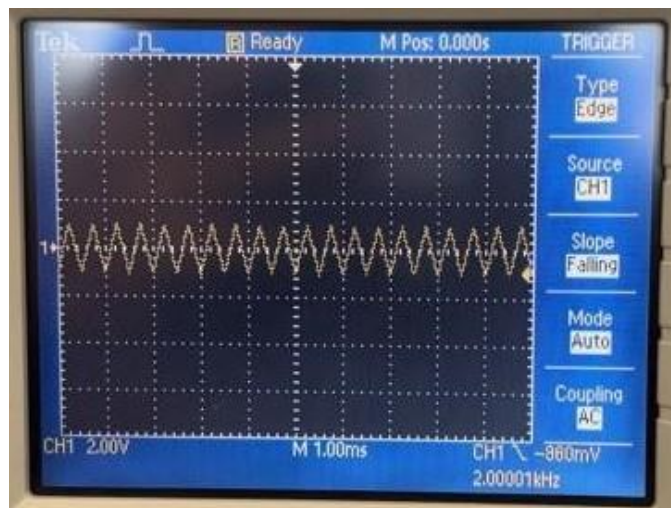


Figure 16: Triggering step 4

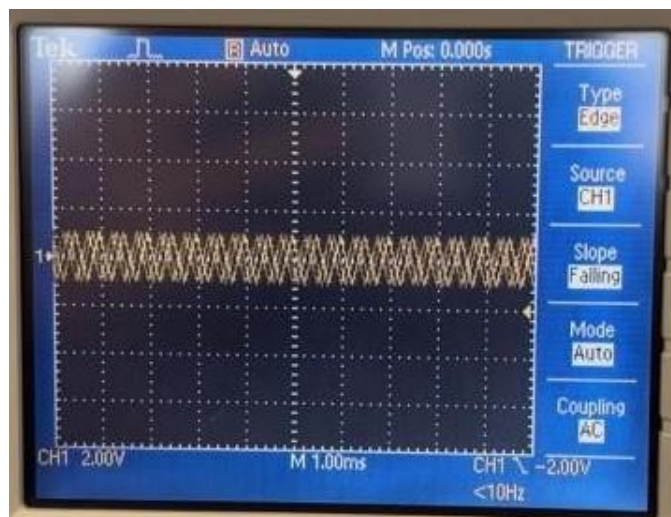


Figure 20: Triggering step 5

As it can be seen from photos, the more triggering arrow gets away from the triggering point, the more the graph gets bad. Therefore, it is important to make correct triggering adjustments while using the oscilloscope.

4. **What is digital to analog converter (DAC)? What is an analog to digital converter (ADC)? What are they used for? Which one is used in oscilloscopes and how? Answer in your report. Apply a 1 Vpp square wave with 5 kHz frequency to the oscilloscope. Then try all the acquisition modes (sample, peak detect, average) and report your observations.**

To illustrate, in electronics the world can be divided into two: analog and digital. Both of them carry information and signals. Moreover, both of them are used even in the daily life. Since their signals are in different forms and since we need both of them, they are need to be converted from one to another. Even in one device, for instance in a computer, there are digital components and analog components. In order for them to communicate, their signals must be converted. These are the reasons why we use digital to analog or analog to digital converters, as known as DAC and ADC. DAC's input digital signals which are consisting of 0's and 1's and converts them in a form that can be used as different voltage or currents in circuits. On the contrary, ADC's take data from the sensors or they take their analog specifications such as voltage, current, etc. as data and converts them into digital signals hold in binary system.

Oscilloscopes are machines that are designed to visualize signals through digital screen. They have electronic components inside them which can be classified as analog. Therefore, they take data from analog components and visualize their data via digital components so they use analog to digital converters, ADC's.

By pressing "ACQUIRE" button, acquire menu is opened. It contains three options: sample, peak detect, average. First of them, sample mode, is recording one point during each waveform interval. Secondly, in the peak detect mode oscilloscope records the maximum and minimum sample points during two waveform intervals and create a whole graph by using them. Lastly, the average mode averages sample points recorded during each waveform interval and graphs them. This allows oscilloscope to reduce the noise.



Figure 21: Amplitude adjustment, 1 Vpp signal will be created

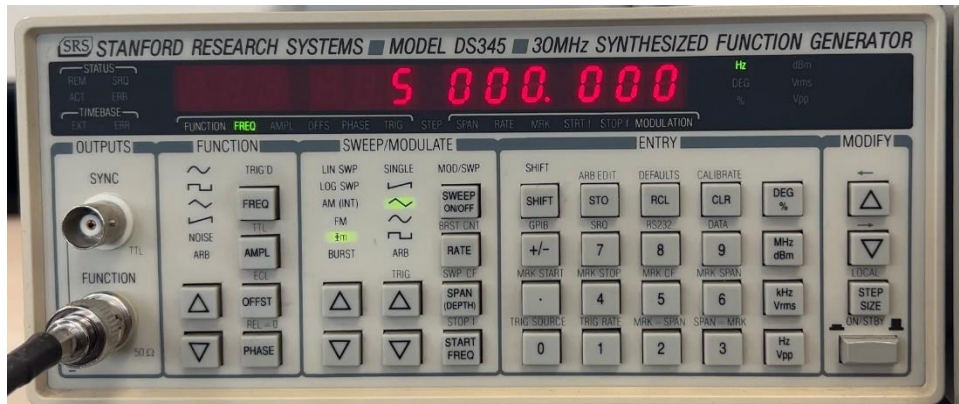


Figure 22: Frequency adjustment

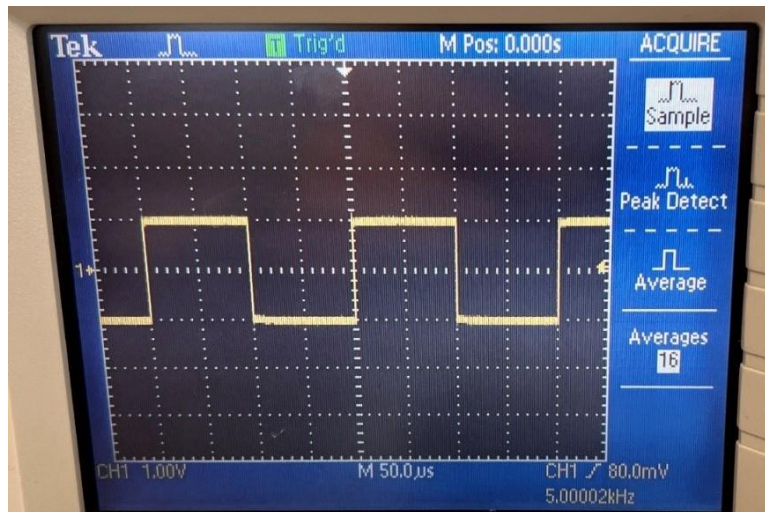


Figure 23: Acquisition, sample mode

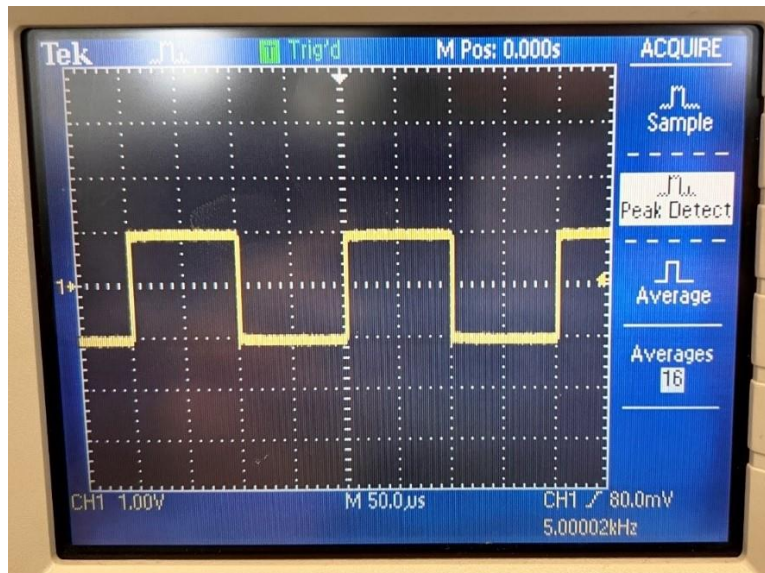


Figure 24: Acquisition, peak detect mode

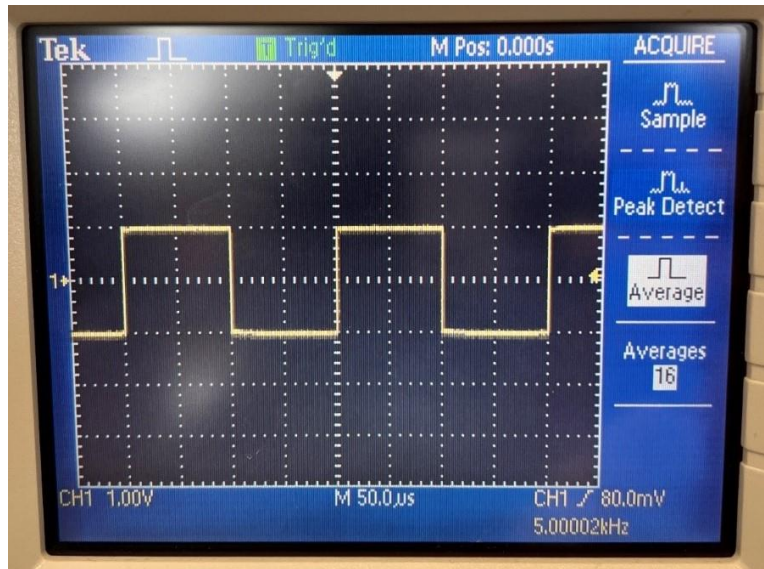


Figure 25: Acquisition, average mode

I observed that in average mode, the lines are thinner, clearer and more stable. In sample mode, again is it thin but the stability is less than the average mode. However, in the peak detect mode the instability and disruption is the most as expected because it only shows the min/max points.

5. Generate a sinusoidal signal with 2 Vpp amplitude and 1 kHz frequency. Also apply a DC offset of 1 V. First, use DC coupling on the oscilloscope and report what you see. Then use AC coupling and report the result. Comment on the difference.



Figure 26: Amplitude adjustment, 2 Vpp signal will be created

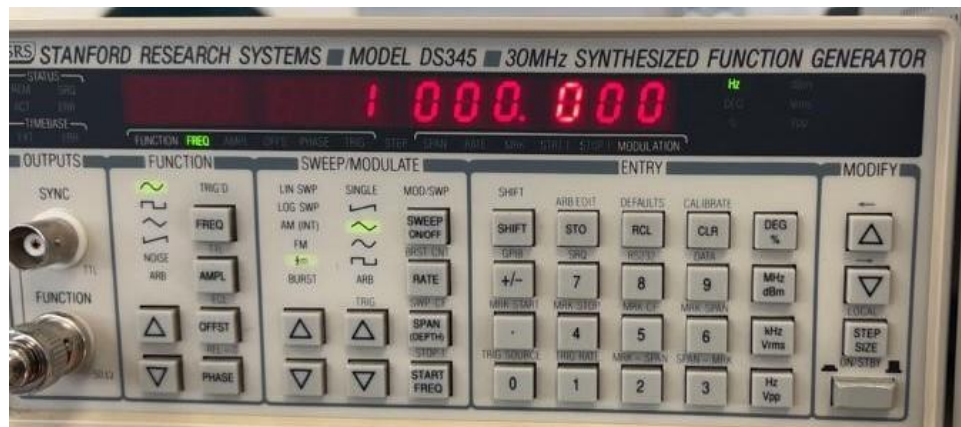


Figure 27: Frequency adjustment

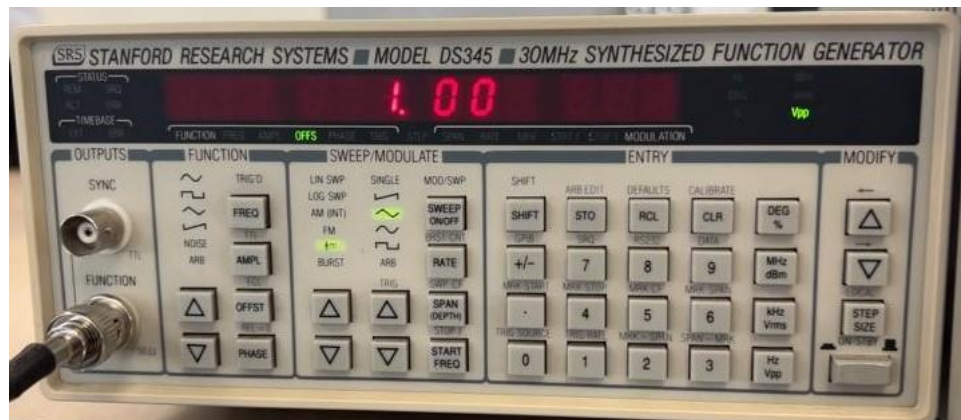


Figure 28: Offset adjustment

By pressing “ACQUIRE” button, acquire menu is opened. To set coupling AC or DC, upper button next to “Coupling” word should be pushed. Photos below are displaying DC and AC couplings, respectively.

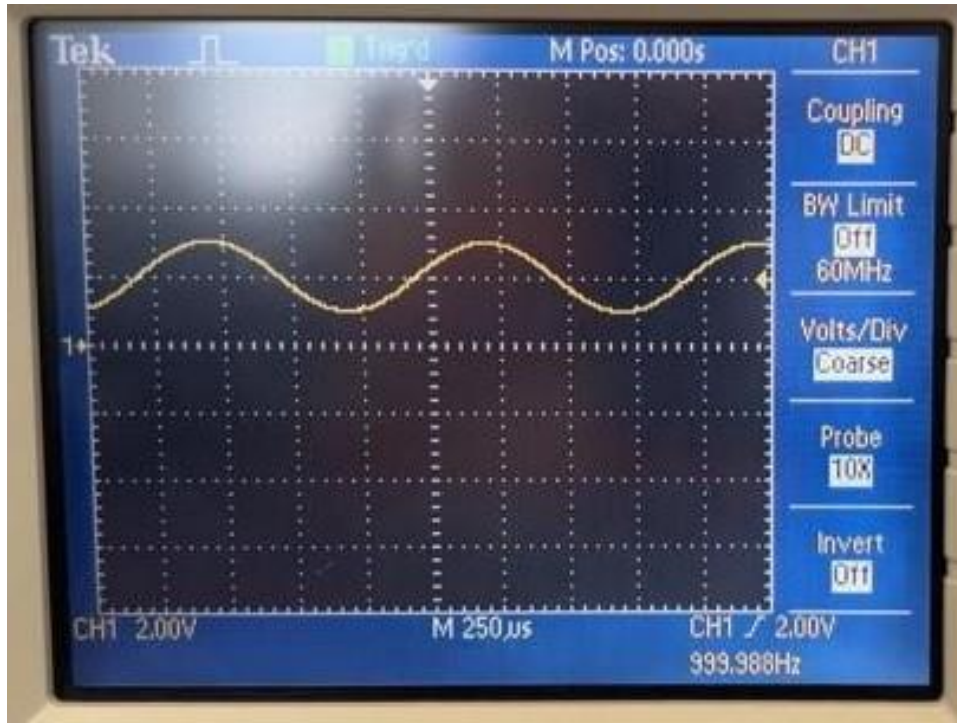


Figure 299: DC coupling

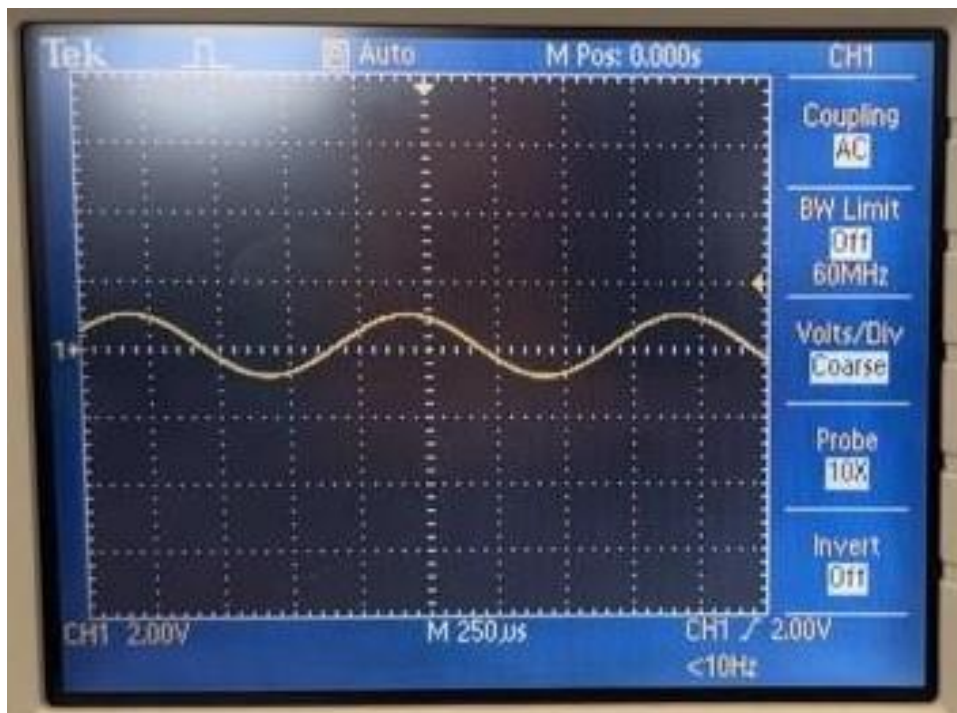


Figure 30: AC coupling

Coupling is a method aims that couple-ing. Its main point is to gather two signals. In AC coupling, oscilloscope does not let DC signals to go through so the visualization is made only by AC signals. They are set onto the middle as it can be seen from the photos above. On the contrary, in DC coupling both signals are passed. And the visualization is consisting of summation of these two. When the signals are applied offset of 1V, roughly this allows to DC voltage difference will be 1V, DC coupling displays the graph above whose middle is not anymore set to 0 level because it is the summation. This is the main point lies behind the location difference relative to 0 level of DC and AC coupling.

6. Check how a breadboard works. Very briefly explain it in your report. Set up the following circuit on your breadboard. Do not forget to connect the grounds. It is common practice to use the long blue lines on your breadboard for ground and red lines for voltage sources as they connect to most other components. Notice that on some breadboards, these lines may be disconnected at the middle of the board.

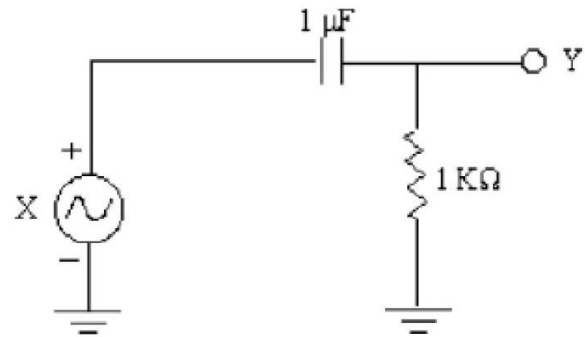


Figure 10: Given circuit design

Apply a 2 Vpp 1 kHz sinusoidal signal (with 0 DC offset) as the X signal. Borrow a friend's probe and display both X and Y signals on the oscilloscope. Make sure you connect the grounds of both channels to the common ground; oscilloscopes measure voltage *differences*. Use channel 1 for X and channel 2 for Y. You should be using channel 1 as the trigger source. Learn how to measure time and voltage differences on an oscilloscope. Measure the delay between signals X and Y. Calculate the phase difference (note that a full period corresponds to a phase of 2π). Then, change the frequency to 100 kHz and repeat. Report (with photos) the results. Try to comment on the difference between different frequencies.

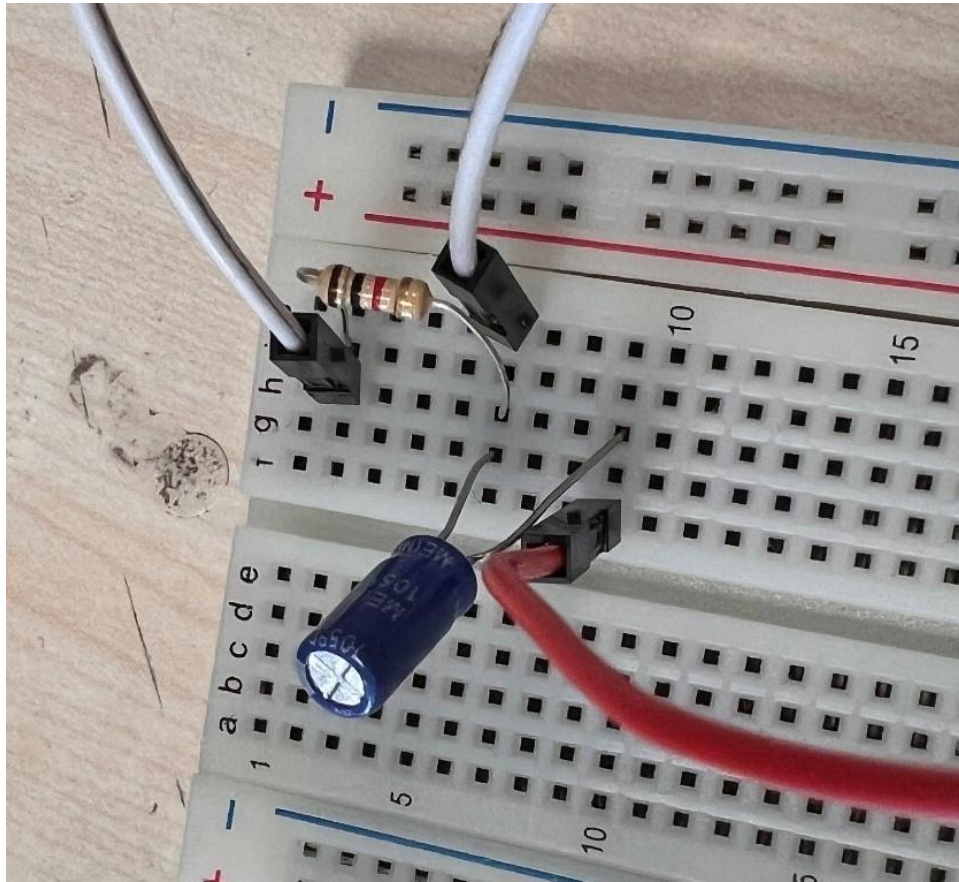


Figure 32: Breadboard connection

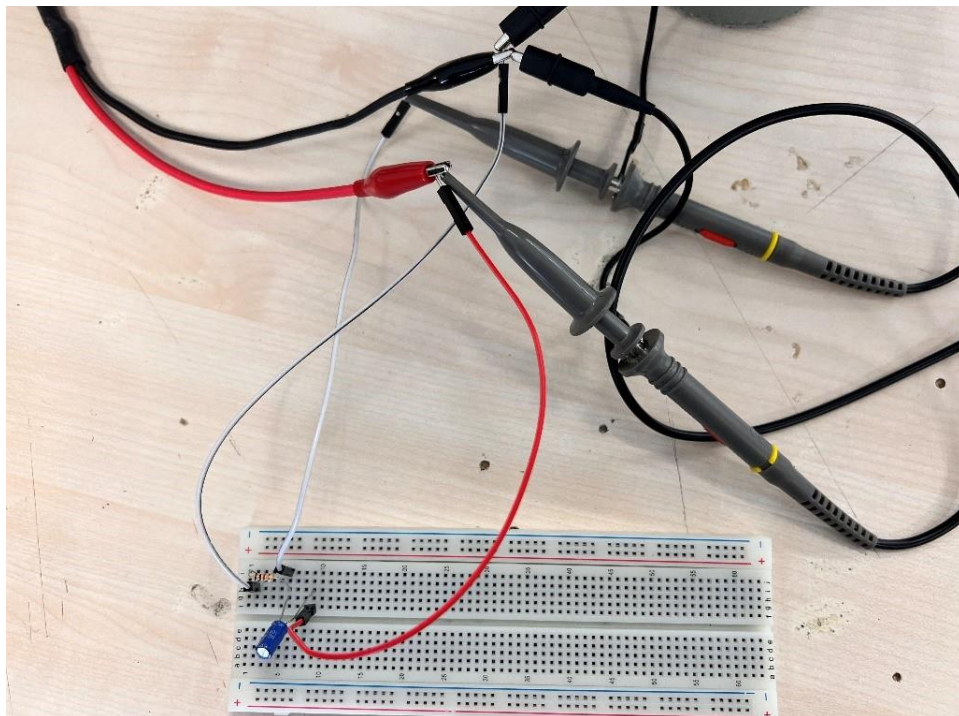


Figure 33: Circuit connections

Firstly, adjusting amplitude to 1 Vpp to create 2 Vpp signals and frequency to 1 kHz. Also displaying the graph, voltage differences and time delay.

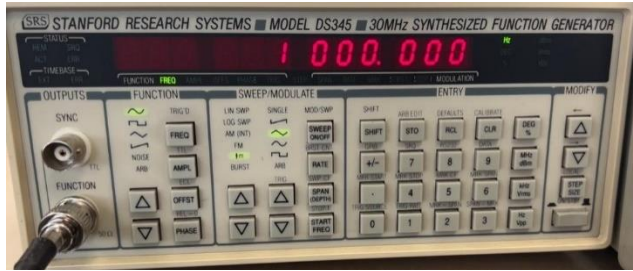


Figure 34: Frequency adjustment

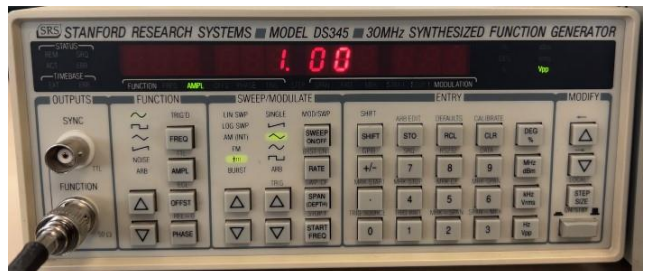


Figure 35: Amplitude adjustment, 2 Vpp signals will be created

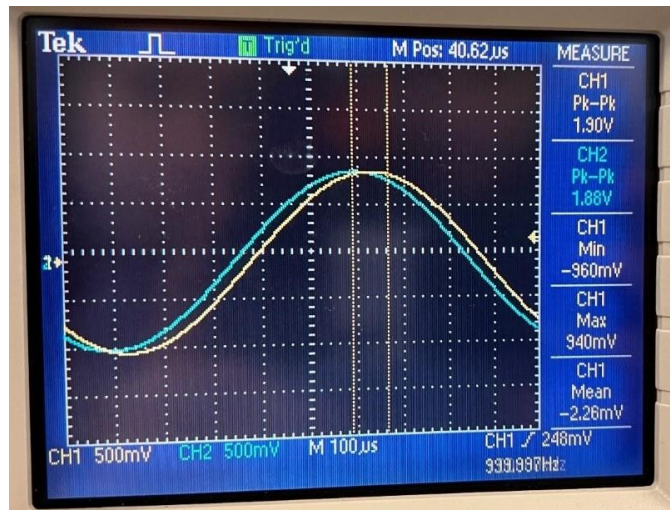


Figure 36: Measurements of 1 kHz signal including pk-pk value

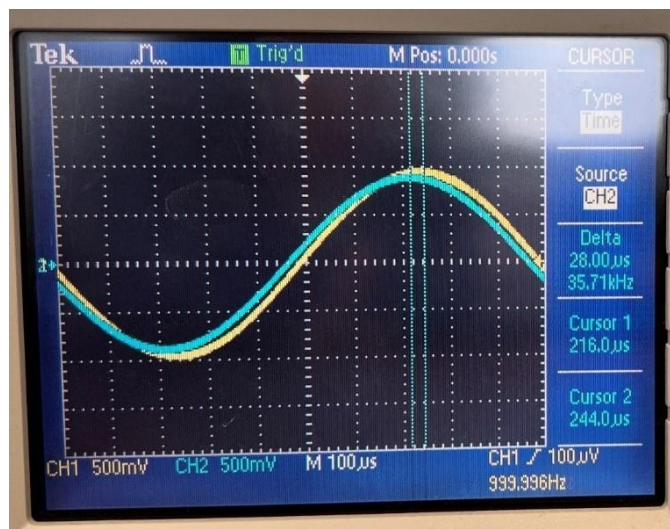


Figure 37: Measurements of 1 kHz signal including time delay

After that, adjusting amplitude to 1 Vpp to create 2 Vpp signals and frequency to 100 kHz.

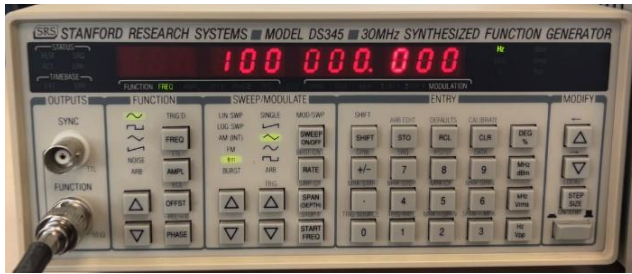


Figure 38: Frequency adjustment



Figure 39: Amplitude adjustment, 2 Vpp signals will be created

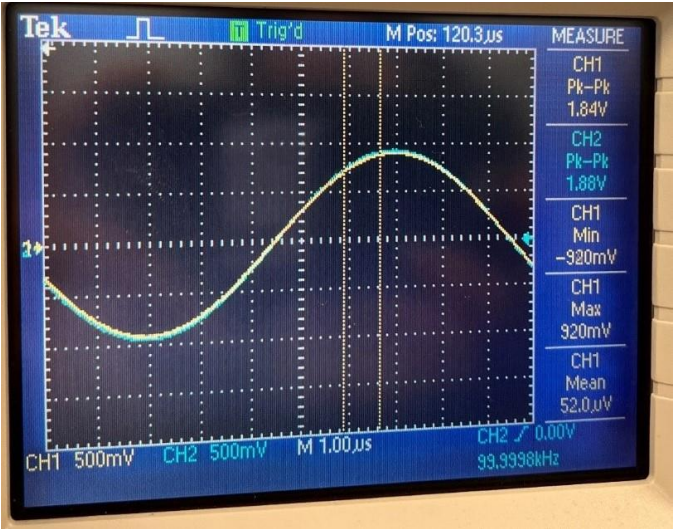


Figure 40: Measurements of 100 kHz signal including pk-pk value

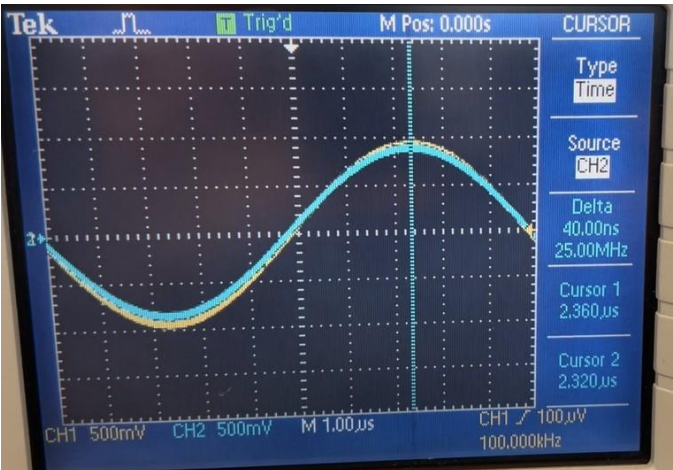


Figure 41: Measurements of 100 kHz signal including time delay

As it can be seen from photos, as frequency gets bigger, graphs get more identical and their time delay gets smaller. This is so because frequency means signals per second, that is, while the first one gets 1000 data per second, second one gets 100*1000. It means that in the second one, data is getting faster and the process is more intense. Therefore, the difference between CH1 and CH2 (or X and Y in this case) is much smaller compared to the first one.

Time delay can be easily obtained by adjusting cursors on to the peaks. When they are adjusted, the difference is shown by oscilloscope under the title “Delta”. In the first case, the time delay is 28 μ s, while, in the second case it is 40 ns. Obviously, first time delay is much bigger than the second, the reason is explained in the previous paragraph.

Phase angle is the angle related to the frequency and time delay. Its’ formula is

$$\varphi = 2\pi \times f \times \Delta t^1$$

where φ stands for phase angle, f stands for frequency and Δt stands for time delay. Therefore, phase angle with frequency of 1 kHz is 0.056π in radians (equivalent to 10.08°) and phase angle with frequency of 100 kHz is 0.008π in radians (1.44°).

Conclusion

The aim of this experiment was to obtain basic knowledge about usage of the oscilloscope and signal generator while introducing basic electronic concepts including circuit design on a breadboard. During experiment, triggering, how to make specific measurements and basic functions of oscilloscope is addressed and studied.

Appendices

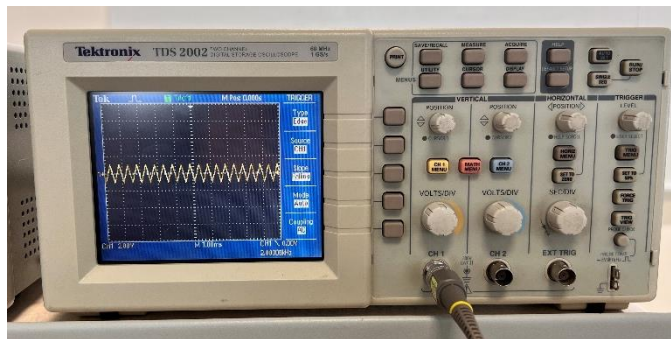


Figure 42: Used oscilloscope, TDS 2002

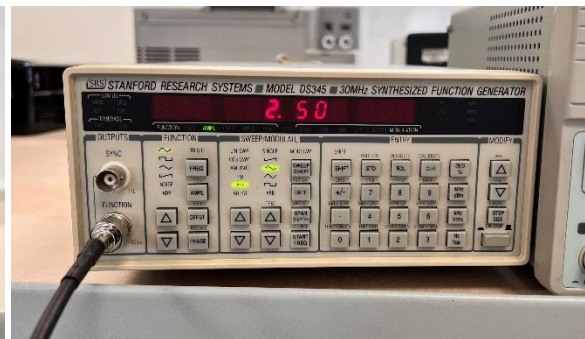


Figure 43: Used signal generator, DS345

¹ <https://voltage-disturbance.com/power-quality/calculating-phase-difference-between-two-waves/>

References

chrome-extension://efaidnbmnnnibpcajpcgglefindmkaj/https://moodle.bilkent.edu.tr/2022-2023-spring/pluginfile.php/36720/mod_resource/content/1/digital_oscilloscope_principles.pdf

<https://voltage-disturbance.com/power-quality/calculating-phase-difference-between-two-waves/>